



TPS22918, 5.5-V, 2-A, 54-mΩ On-Resistance Load Switch

1 Features

- Integrated Single Channel Load Switch
- Ambient Operating Temperature: -40°C to $+105^{\circ}\text{C}$
- Input Voltage Range: 1 V to 5.5 V
- Low On-Resistance (R_{ON})
 - $R_{\text{ON}} = 53\text{ m}\Omega$ (typ) at $V_{\text{IN}} = 5\text{ V}$
 - $R_{\text{ON}} = 54\text{ m}\Omega$ (typ) at $V_{\text{IN}} = 3.3\text{ V}$
- 2-A Maximum Continuous Switch Current
- Low Quiescent Current
 - $8.7\text{ }\mu\text{A}$ (typ) at $V_{\text{IN}} = 3.3\text{ V}$
- Low Control Input Threshold Enables Use of 1 V or Higher GPIO
- Adjustable Quick Output Discharge (QOD)
- Configurable Rise Time With CT Pin
- Small SOT23-6 Package
 - 2.9-mm $\times$ 2.8-mm, 0.95-mm Pitch, 1.45-mm Height (DBV)
- ESD Performance Tested per JESD 22
 - $\pm 2\text{ kV}$ HBM and $\pm 1\text{ kV}$ CDM

2 Applications

- Industrial Systems
- Set Top Box
- Blood Glucose Meters
- Electronic Point of Sale

3 Description

The TPS22918 is a small, low R_{ON} , single channel load switch with both configurable rise time and configurable quick output discharge. The device contains an N-channel MOSFET that can operate over an input voltage range of 1 V to 5.5 V and can support a maximum continuous current of 2-A. The switch is controlled by an on and off input, which is capable of interfacing directly with low-voltage control signals.

The configurable rise time of the device greatly reduces inrush current caused by large bulk load capacitances, thereby reducing or eliminating power supply droop. The TPS22918 features a quick output discharge (QOD) pin, which controls the fall time of the device to allow design flexibility for power down and sequencing.

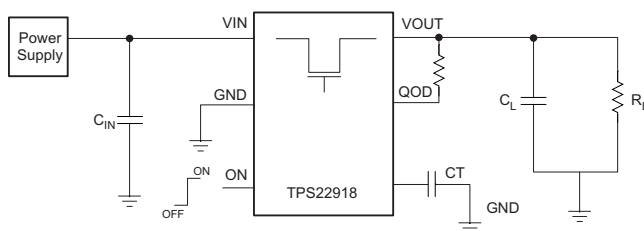
The TPS22918 is available in a small, leaded SOT-23 package (DBV) which allows to visually inspect solder joints. The device is characterized for operation over the free-air temperature range of -40°C to $+105^{\circ}\text{C}$.

Device Information ⁽¹⁾

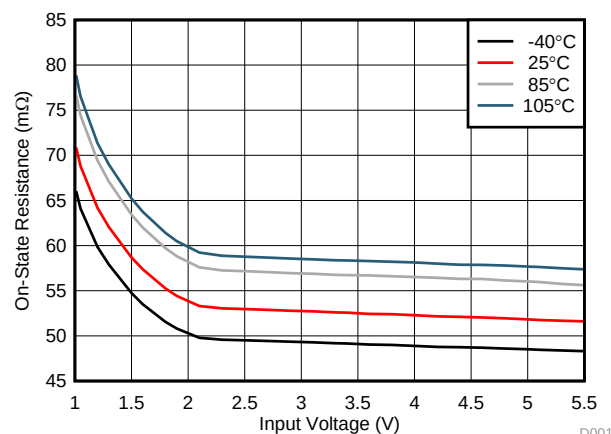
PART NUMBER	PACKAGE	BODY SIZE (NOM)
TPS22918	SOT-23 (6)	2.90 mm $\times$ 2.80 mm

(1) For all available packages, see the orderable addendum at the end of the datasheet.

Simplified Schematic



R_{ON} vs V_{IN} ($I_{\text{OUT}} = -200\text{ mA}$)



D001



4 Device and Documentation Support

4.1 Documentation Support

4.1.1 Related Documentation

For related documentation see the following:

TPS22918 5.5-V, 2-A, 50-mΩ On-Resistance Load Switch Evaluation Module , [SLVUAP0](#)

4.2 Community Resources

The following links connect to TI community resources. Linked contents are provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's [Terms of Use](#).

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Design Support *TI's Design Support* Quickly find helpful E2E forums along with design support tools and contact information for technical support.

4.3 Trademarks

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4.4 Electrostatic Discharge Caution



These devices have limited built-in ESD protection. The leads should be shorted together or the device placed in conductive foam during storage or handling to prevent electrostatic damage to the MOS gates.

4.5 Glossary

[SLYZ022](#) — *TI Glossary*.

This glossary lists and explains terms, acronyms, and definitions.

5 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

PACKAGING INFORMATION

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead/Ball Finish (6)	MSL Peak Temp (3)	Op Temp (°C)	Device Marking (4/5)	Samples
TPS22918DBVR	PREVIEW	SOT-23	DBV	6	3000	Green (RoHS & no Sb/Br)	CU SN	Level-2-260C-1 YEAR	-40 to 105	13MW	
TPS22918DBVT	PREVIEW	SOT-23	DBV	6	250	Green (RoHS & no Sb/Br)	CU SN	Level-2-260C-1 YEAR	-40 to 105	13MW	

(1) The marketing status values are defined as follows:

ACTIVE: Product device recommended for new designs.

LIFEBUY: TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.

NRND: Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.

PREVIEW: Device has been announced but is not in production. Samples may or may not be available.

OBSOLETE: TI has discontinued the production of the device.

(2) Eco Plan - The planned eco-friendly classification: Pb-Free (RoHS), Pb-Free (RoHS Exempt), or Green (RoHS & no Sb/Br) - please check <http://www.ti.com/productcontent> for the latest availability information and additional product content details.

TBD: The Pb-Free/Green conversion plan has not been defined.

Pb-Free (RoHS): TI's terms "Lead-Free" or "Pb-Free" mean semiconductor products that are compatible with the current RoHS requirements for all 6 substances, including the requirement that lead not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, TI Pb-Free products are suitable for use in specified lead-free processes.

Pb-Free (RoHS Exempt): This component has a RoHS exemption for either 1) lead-based flip-chip solder bumps used between the die and package, or 2) lead-based die adhesive used between the die and leadframe. The component is otherwise considered Pb-Free (RoHS compatible) as defined above.

Green (RoHS & no Sb/Br): TI defines "Green" to mean Pb-Free (RoHS compatible), and free of Bromine (Br) and Antimony (Sb) based flame retardants (Br or Sb do not exceed 0.1% by weight in homogeneous material)

(3) MSL, Peak Temp. - The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.

(4) There may be additional marking, which relates to the logo, the lot trace code information, or the environmental category on the device.

(5) Multiple Device Markings will be inside parentheses. Only one Device Marking contained in parentheses and separated by a "-" will appear on a device. If a line is indented then it is a continuation of the previous line and the two combined represent the entire Device Marking for that device.

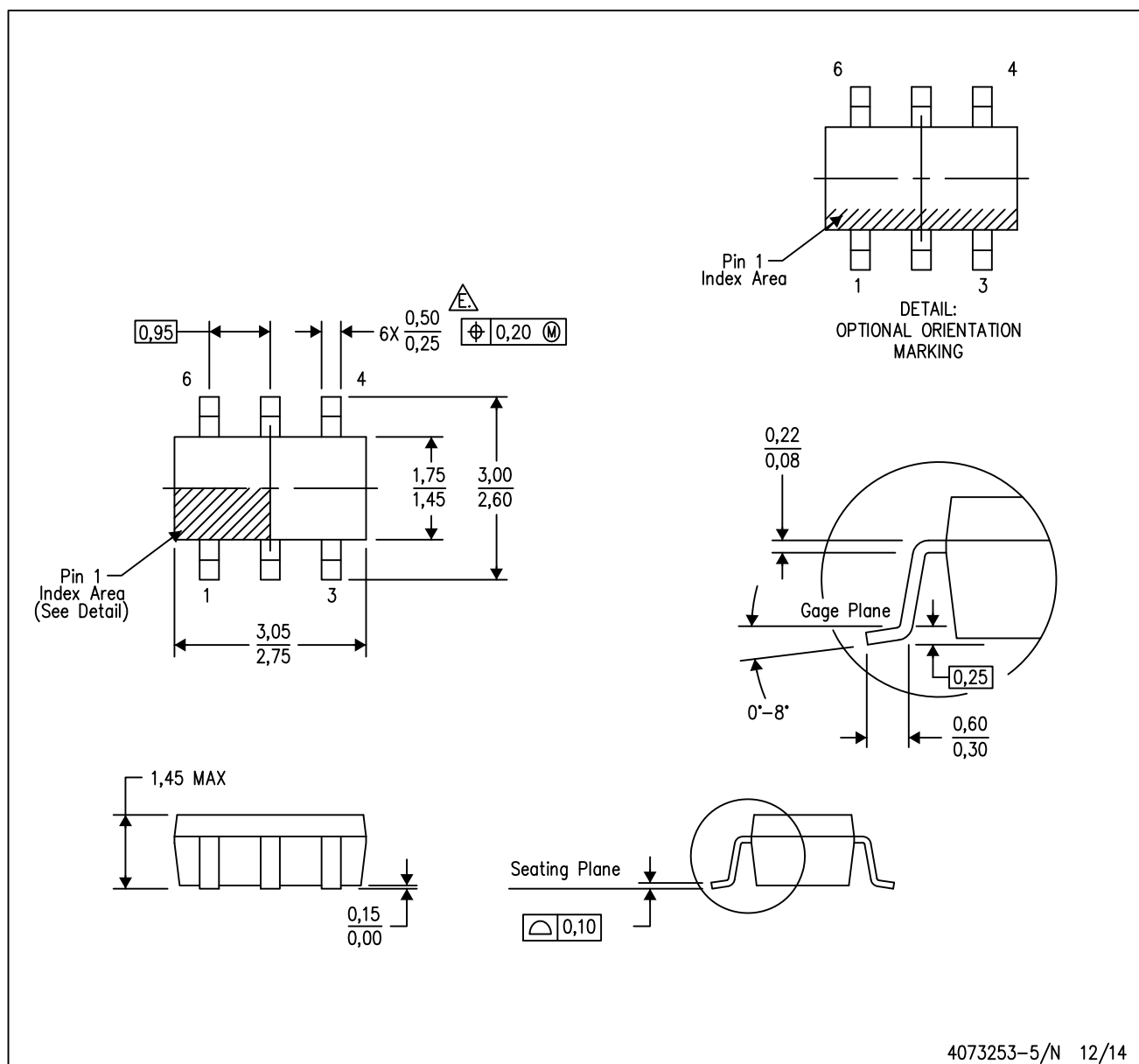
(6) Lead/Ball Finish - Orderable Devices may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead/Ball Finish values may wrap to two lines if the finish value exceeds the maximum column width.

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DBV (R-PDSO-G6)

PLASTIC SMALL-OUTLINE PACKAGE

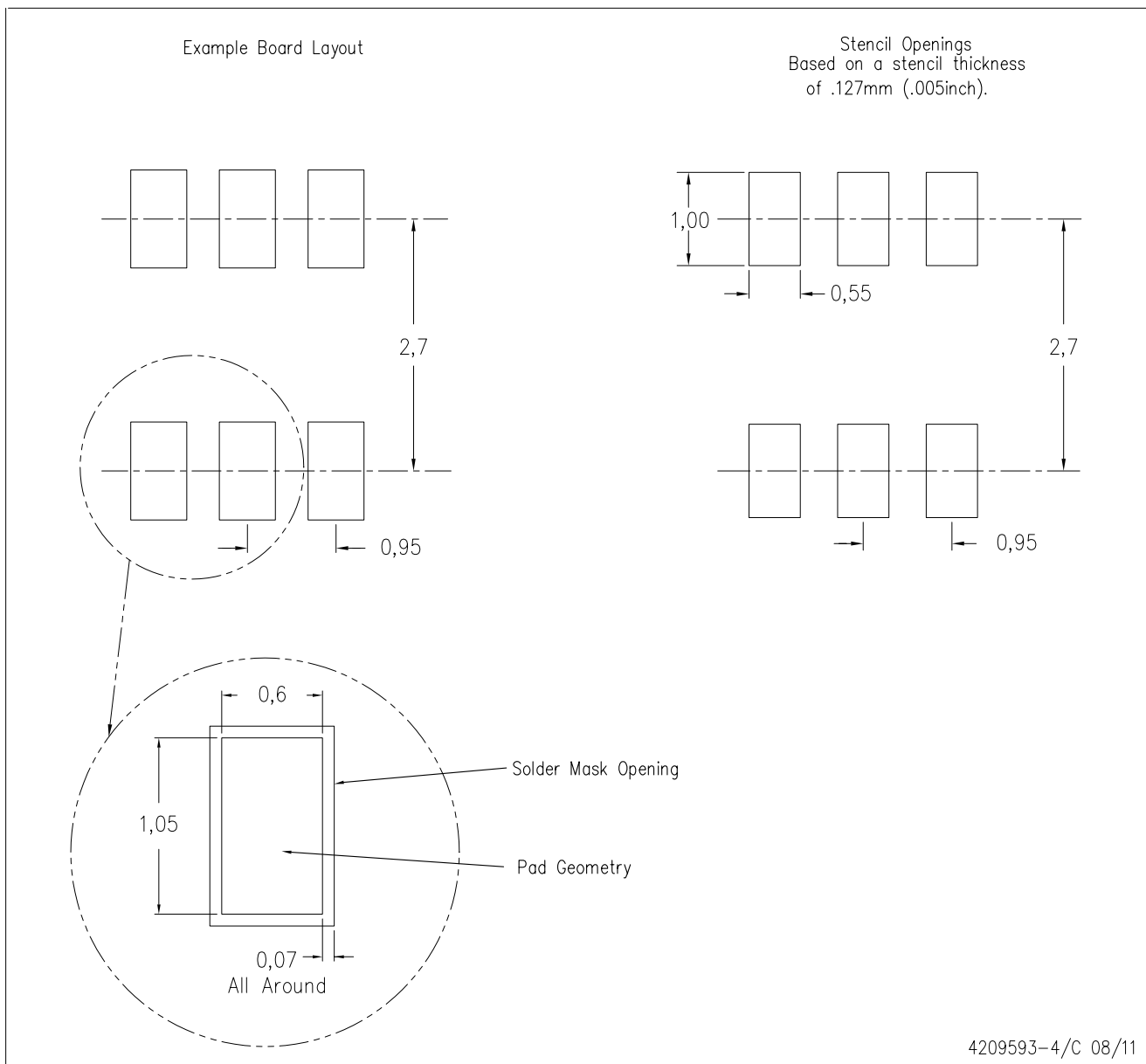


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- NOTES:
- A. All linear dimensions are in millimeters.
 - B. This drawing is subject to change without notice.
 - C. Body dimensions do not include mold flash or protrusion. Mold flash and protrusion shall not exceed 0.15 per side.
 - D. Leads 1,2,3 may be wider than leads 4,5,6 for package orientation.
 - E. Falls within JEDEC MO-178 Variation AB, except minimum lead width.

DBV (R-PDSO-G6)

PLASTIC SMALL OUTLINE



- NOTES:
- A. All linear dimensions are in millimeters.
 - B. This drawing is subject to change without notice.
 - C. Customers should place a note on the circuit board fabrication drawing not to alter the center solder mask defined pad.
 - D. Publication IPC-7351 is recommended for alternate designs.
 - E. Laser cutting apertures with trapezoidal walls and also rounding corners will offer better paste release. Customers should contact their board assembly site for stencil design recommendations. Example stencil design based on a 50% volumetric metal load solder paste. Refer to IPC-7525 for other stencil recommendations.

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